

Automation Project Proposal

Course Companion — AI Knowledge Checks and Weak-Spot Tracking for the Growth Portal

Ad-hoc Activity Plan, Item 4 · Prepared by Leander · 2026-08-24

Problem

Growth Portal courses today are watch-only. When someone finishes a course, the only signals are a progress percentage and the "I learned something useful" button. Nothing verifies that the viewer actually **understood** the material, and nothing checks whether it **stuck** a week later. For the learner that means passive watching with no feedback; for mentors and EMs it means course completion counts that say little about what was actually learned.

Proposed project

A **Course Companion** that closes the learning loop on every course, using AI where it is strong (generating good questions from a transcript) and keeping a human in the loop where it matters (approving them):

Step	What happens
1. Generate	A short knowledge check (3–5 questions) is generated from the course transcript using Claude. It stays a draft until the course author or an EM approves it.
2. Answer	The learner takes the check when they finish the course.
3. Score	A comprehension score is recorded alongside the existing completion signal.
4. Track	Missed questions land on the learner's personal weak-spot list , tagged by course and concept.
5. Resurface	Weak spots come back later as a short spaced retest — so the result reflects retention, not just same-day recall.

Why this project

This is the loop my 2-week training plan ran manually, and it worked: a quiz before each build session, every miss logged to a running weak-spot list, and the final retest prepared by drilling that list instead of re-reading everything. The Day 10 retest answers include a question-by-question comparison against the Day 1 baseline. The proposal is simply to automate the mechanism that already produced that result, and make it available on every Growth Portal course — which covers both examples in the activity item: an AI learning activity that speeds up learning, delivered as a GP integration.

Phased scope

Phase	Scope	Delivers
MVP	Standalone service alongside GP — no GP changes needed. Takes a course transcript, generates a reviewable quiz, serves it, stores scores and the weak-spot list.	A working end-to-end demo on one real course, enough to judge question quality and the review workflow.
Phase 1.5	GP integration: the knowledge check appears at course completion; the comprehension score shows on the course card next to progress %.	The loop becomes part of the normal course experience.
Phase 2	Mentor/EM view: weak-spot rollups per mentee, plus baseline and retest generation for individual training plans.	Automates the assessment side of training plans that is run by hand today.

Risks and mitigations

- **Question quality** — AI-generated quizzes can be wrong or trivial. Every quiz is a draft until a human (course author or EM) approves it.
- **How the data is perceived** — scores could feel like performance rating. The Companion is scoped as a **learning signal, not a performance-rating input**: the weak-spot list belongs to the learner, and mentor rollups (phase 2) are mentorship-oriented.
- **Missing transcripts** — courses without a transcript get one auto-generated (speech-to-text) during ingestion. Side benefit: transcripts also make the course library searchable.

Proposed next step

If the direction looks right, I would start with the MVP — a small NestJS service using the Claude API, matching the team stack — and demo it on one existing GP course. I estimate the MVP as roughly a one-week time-box at the same daily hours as the course work; I will propose a concrete schedule once the phase-1 scope is agreed.